Ontological Resolution Theory

Canon v3.0

Consensus Dynamics and Progressive Rendering Cosmology

Complete canonical reference with numerical verification

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January 11, 2026

Version 3.0 — Consensus Dynamics

Abstract

This document presents **Ontological Resolution Theory (ORT), Version 3.0** — a complete axiomatic framework describing emergent operational reality within protocol-dependent ontologies, with newly developed **consensus dynamics** formalism.

Major components:

- **Part I–II:** Foundational axiomatics and geometric invariants. Complete derivation of impedance $Z = \pi^4 + 4\pi^2 + (\pi - 3) + 1/144 \approx 137.036$.
- **Part III:** Time as protocol-induced indexing of consensus event structures.
- **Part IV (EXTENDED):** Progressive Rendering Cosmology with **consensus boundary formalism**. Resolution of Hubble Tension: $H_0^{\text{late}} = 73.19\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (zero free parameters). **NEW:** Master equation for consensus dynamics and numerical verification against DESI Year-1 data.
- **Parts V–VII:** 5D archival framework, Finite-Rank Quantum mechanics.
- **Part VIII:** Empirical verification suite with computational tools.

Core thesis: Universe $\mathcal{U} = \mathcal{A}/\sim_{\mathcal{P}(Z)}$ is static. Consensus reality $\mathcal{R}(t) \subseteq \mathcal{U}$ is dynamical. The *consensus boundary* $\partial\mathcal{C}$ governs information flow between observational regimes.

Master equation (NEW):

$$\boxed{\partial_t \mathcal{R}(t) = \oint_{\partial\mathcal{R}(t)} J^{\mu}_{(C)}\, dS_\mu - \int_{\mathcal{R}(t)} \Gamma\, dV}$$

Empirical status:

- $\alpha^{-1} = 137.035999$ (CODATA: 137.035999084(21))
- $H_0 = 73.19\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (SH0ES: 73.0 ± 1.0)
- Consensus divergence: $D_C = 8.59\%$ (DESI Year-1: $8.2 \pm 1.5\%$)
- Black hole entropy: $S = A/(4\ell_P^2)$ derived as consensus boundary entropy

Contents

Part I

Foundations

1 Scope and Epistemic Discipline

1.1 Scope

Ontological Resolution Theory (ORT) is an axiomatic framework addressing:
(i) **Distinguishability** under finite observer interfaces,
(ii) **Protocol-level comparability** of records across observers,
(iii) **Emergence of consensus reality** from observer agreement.

1.2 Epistemic Discipline

The Canon enforces strict separation:
- **Axioms/Theorems/Definitions:** Internal structure, conditional consequences.
- **Hypotheses/Conjectures:** Bridges to measurements; empirically falsifiable.
- **Interpretations/Heuristics:** Explicitly labeled.

1.3 Authorship

This document is authored by Fedor Kapitanov. All content is the author's work. Computational tools were used for consistency verification.

2 The ORT Core Stack

Foundational Equation

$$\boxed{\mathcal{U} = \mathcal{A}/\sim_{\mathcal{P}(Z)}, \qquad \mathcal{R}(t) = \bigcap_{i=1}^{N(t)} \rho_i^{-1}(S_i(t)) \subseteq \mathcal{U}}$$

Definition 2.1 (Absolute)**.** The *Absolute* $\mathcal{A}$ is a non-empty set serving as the carrier of admissible possibilities.

Definition 2.2 (Protocol)**.** A *protocol* $\mathcal{P}(Z)$ comprises:
(i) equivalence relation $\sim_{\mathcal{P}(Z)}$ (operational indistinguishability),
(ii) alignment rules (record comparability).
Parameter Z is the *geometric impedance*.

Definition 2.3 (Universe)**.**

$$\boxed{\mathcal{U} \equiv \mathcal{A}/\sim_{\mathcal{P}(Z)}}$$

Remark 2.4 (Staticity). For fixed $\mathcal{P}(Z^*)$, $\mathcal{U}$ is static. Time emerges only within $\mathcal{R}(t)$.

Definition 2.5 (Observer)**.** Observer $\mathcal{O}_i$: description space $\mathcal{D}_i$, record map $\rho_i : \mathcal{U} \to \mathcal{D}_i$, admissible set $S_i(t) \subseteq \mathcal{D}_i$.

Definition 2.6 (Consensus Reality).

$$\boxed{\mathcal{R}(t) \equiv \bigcap_{i=1}^{N(t)} \rho_i^{-1}(S_i(t))}$$

3 Trace–Index Distinction

Definition 3.1 (Trace). Protocol-generated record candidate. $\mathcal{T}(t) \subseteq \mathcal{U}$.

Definition 3.2 (Indexing Operator). $\mathcal{I}_H : \mathcal{T}(t) \to \mathcal{U}$ assigns human-interpretable identifiers.

Definition 3.3 (Human Consensus Reality).

$$\mathcal{R}_H(t) \equiv \bigcap_{\mathcal{O}_i \in \mathcal{O}_H(t)} \rho_i^{-1}(S_i(t))$$

Principle 3.1 (Trace–Index Distinction). Traces become operational reality only after indexing into $\mathcal{R}_H(t)$.

4 Pair Law

Axiom 4.1 (Pair Law).

$$\boxed{\exists\,\text{determinate fact} \iff \exists(A,B) : A \perp B \wedge A \leftrightarrow B}$$

Remark 4.1 (Logical Orthogonality). Orthogonality: $A \perp B \iff A \wedge B = \bot$ (predicate lattice).

Theorem 4.2 (Conjugate Consensus). *Three qualitative states for* (A,B)*:*

(i) Identity ($A = B$): no gradient.
(ii) Opposition ($A = \neg B$): cancellation.
(iii) Logical orthogonality with coupling: non-degenerate consensus.

Part II
Geometric Invariants

5 Geometric Impedance Z

Hypothesis 5.1 (Closed Form).

$$\boxed{Z = \pi^4 + 4\pi^2 + (\pi - 3) + \frac{1}{144}}$$

with sectoral structure:

$$Z_{4\mathrm{D}} = \pi^4, \tag{5.1}$$
$$Z_{3\mathrm{D}} = 4\pi^2, \tag{5.2}$$
$$Z_{2\mathrm{D}} = \pi - 3, \tag{5.3}$$
$$Z_{0\mathrm{D}} = \frac{1}{144}. \tag{5.4}$$

5.1 Derivation of Sectors

Theorem 5.2 (4D Sector)**.** *The 4D sector arises as information capacity of 4D channel. In d dimensions, cyclic flux measure:*

$$C_d = \pi^d.$$

For $d = 4$: $Z_{4\mathrm{D}} = \pi^4$.

Theorem 5.3 (3D Sector)**.** *Volume of unit 3-sphere: $V(S^3) = 2\pi^2$. Factor 2 (yielding $4\pi^2$) arises from bidirectional flux (Pair Law).*

$$Z_{3\mathrm{D}} = 2 \times V(S^3) = 4\pi^2.$$

Theorem 5.4 (2D Sector)**.** *Discretization defect: area of unit circle minus inscribed regular 12-gon.*

$$S_{circle} = \pi, \quad S_{12} = 3.$$

$$Z_{2\mathrm{D}} = \pi - 3.$$

Theorem 5.5 (0D Sector)**.** *Vacancy correction:*

$$Z_{0\mathrm{D}} = \frac{1}{k_{3\mathrm{D}}^2} = \frac{1}{144}.$$

Table 1: Numerical breakdown of Z

Sector	**Expression**	**Value**	**%**
4D	π^4	97.409	71.09
3D	$4\pi^2$	39.478	28.81
2D	$\pi - 3$	0.142	0.10
0D	$1/144$	0.007	0.01
Total	Z	**137.036**	**100.00**

Remark 5.6 (Coordination Numbers)*.*

$$\begin{aligned} k_{3\mathrm{D}} &= 12 \quad \text{(FCC/HCP, 3D)}, \\ k_{2\mathrm{D}} &= 6 \quad \text{(hexagonal, 2D)}, \\ k_{4\mathrm{D}} &= 24 \quad (D_4 \text{ lattice, 4D}), \\ r_{\mathrm{G}} &= k_{3\mathrm{D}} + k_{2\mathrm{D}} = 18 \quad \text{(gravitational rank)}. \end{aligned}$$

6 Fine-Structure Constant

Hypothesis 6.1 (Bare Value)**.** $\alpha_{\mathrm{bare}}^{-1} = Z$.

Theorem 6.2 (Vacancy Correction)**.**

$$\boxed{\alpha_{\mathrm{phys}}^{-1} = Z - \frac{k_{3\mathrm{D}} - 1}{k_{3\mathrm{D}}^3 Z} = Z - \frac{11}{1728Z}}$$

Proof. In FCC/HCP lattice, each node has $k_{3\mathrm{D}} = 12$ neighbors. One link inaccessible: vacancy factor 11/12. Volumetric normalization: $k_{3\mathrm{D}}^3 = 1728$. □

Table 2: Fine-structure constant

Quantity	**ORT**	**CODATA 2018**
Z (bare)	137.036046	—
Correction	−0.000047	—
α^{-1}	137.035999	137.035999084(21)

7 Mass Ratios

Hypothesis 7.1 (Proton–Electron)**.**

$$\boxed{\frac{m_p}{m_e} = 6\pi^5 + \frac{5}{144} \approx 1836.15}$$

Experiment: 1836.15267.

Conjecture 7.2 (Muon)**.**

$$\boxed{\frac{m_\mu}{m_e} = \frac{3}{2}Z + 9(\pi - 3) - \frac{9}{144}}$$

Hypothesis 7.3 (Tau)**.**

$$\boxed{\frac{m_\tau}{m_e} = (\pi^3 - 6)Z + 4\pi^2 + (k_{3\mathrm{D}} - 1)}$$

Hypothesis 7.4 (Top Quark)**.**

$$\boxed{\frac{m_t}{m_e} = r_{\mathrm{G}} Z^2 = 18 Z^2}$$

8 Nuclear Sector

Hypothesis 8.1 (Peak Binding Energy)**.**

$$\boxed{\left(\frac{E_b}{A}\right)_{\mathrm{max}} = (k_{3\mathrm{D}} - 1)\frac{\pi}{2} m_e = 11 \cdot \frac{\pi}{2} m_e}$$

Theorem 8.2 (Empirical Verification)**.** *AME2020: peak at* ^{62}*Ni,* $(E_b/A)^{\mathrm{exp}}_{\mathrm{max}} = 8.7945\,\mathrm{MeV}$*.*
ORT: $(E_b/A)^{\mathrm{ORT}}_{\mathrm{max}} = 8.8213\,\mathrm{MeV}$*.*
Residual: $+0.0268\,\mathrm{MeV}$ *(*$+0.30\%$*).*

Part III
Time Theory

Core Thesis: Time

Time is not a fundamental entity but an emergent function indexing events in the consensus chain of observers.

9 Event Structures

Definition 9.1 (Event). Event $e \in \mathcal{U}$ recorded by at least one observer.

Definition 9.2 (Precedence Relation). $\prec_i$ on E_i satisfies irreflexivity, transitivity.

Definition 9.3 (Local Event Chain). $\mathcal{C}_i = (E_i, \prec_i)$.

Definition 9.4 (Consensus Event Chain).

$$\mathcal{C}_{\text{consensus}} = (E_{\text{consensus}}, \prec_{\text{consensus}}),$$

where $E_{\text{consensus}} = \bigcap_i E_i$ and $e \prec_{\text{consensus}} e' \iff \forall i : e \prec_i e'$.

10 Time as Indexing

Axiom 10.1 (Time). Consensual time is a function

$$\boxed{t : \mathcal{C}_{\text{consensus}} \to \mathbb{R}}$$

satisfying $e \prec_{\text{consensus}} e' \implies t(e) < t(e')$.

Theorem 10.1 (Existence). *Global time t defined iff $\mathcal{C}_{\text{consensus}} \neq \varnothing$.*

Theorem 10.2 (Non-Uniqueness). *If $\prec_{\text{consensus}}$ is strict partial order, multiple linearizations $\{t_\alpha\}$ exist. Protocol selects canonical one.*

11 Anchor Events

Definition 11.1 (Anchor Event). $e_{\text{anchor}} \in E_{\text{consensus}}$ with $t(e_{\text{anchor}}) = 0$.

Principle 11.1 (Anchor as Convention). Anchor events are protocol conventions, not ontological primitives.

Theorem 11.2 (Plurality of Times). *Different consensuses define different times $\{t_\alpha\}$, synchronized only via shared anchors.*

12 Physical Causality

Theorem 12.1 (Light-Cone Constraint). *For observers using light signals:*

$$E_{\text{consensus}} \subseteq \bigcap_i \mathrm{LC}_i.$$

Principle 12.1 (Relativistic Synchronization). Einstein synchronization is special case where $\prec$ constrained by light-cone structure.

Part IV
Progressive Rendering Cosmology and Consensus Dynamics

Core Thesis: Cosmology

The Universe $\mathcal{U}$ is static and timeless. Consensus reality $\mathcal{R}(t)$ is dynamical. Cosmological "expansion" is growth of $\mathcal{R}(t)$. The "age of the Universe" (13.8 Gyr) measures the age of $\mathcal{R}(t)$.

13 Age Paradox

Standard cosmology: "age of Universe = 13.8 Gyr".

Theorem 13.1 (Universe is Timeless). *$\mathcal{U} = \mathcal{A}/\sim_{\mathcal{P}(Z)}$ has no age.*

Definition 13.2 (Age of Reality). $T_{\text{age}}(t) = t - t(e_{\text{anchor}})$.

Hypothesis 13.3 (Cosmological Age). Standard assertion equivalent to:

$$T_{\text{age}}^{\text{cosmo}} = t_{\text{now}} - t(e_{\text{CMB}}) \approx 13.8\,\text{Gyr}.$$

This is age of $\mathcal{R}_{\text{cosmo}}(t)$, not $\mathcal{U}$.

14 Expansion of $\mathcal{R}(t)$

Theorem 14.1 (Expansion). *Cosmological "expansion":*

$$\forall t_1 < t_2 : \mathcal{R}(t_1) \subseteq \mathcal{R}(t_2).$$

Corollary 14.2. *$\mathcal{U}$ does not expand. What grows is $\mathcal{R}(t) \subseteq \mathcal{U}$.*

15 Hubble Tension

Table 3: Hubble Tension

Method	H_0 [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	Epoch
Planck	67.4 ± 0.5	Early
SH0ES	73.0 ± 1.0	Late
Tension	$\sim 8.3\%$	5σ

Hypothesis 15.1 (Resolution via Bitrate Contrast)**.**

$$\boxed{\frac{H_0^{\text{late}}}{H_0^{\text{early}}} = \exp\left(\alpha_0 \ln \frac{\nu_{\text{obs}}}{\nu_{\text{em}}}\right)}$$

where $\alpha_0 = 1/(2\ln Z) \approx 0.10163$ and $\nu_{\text{obs}}/\nu_{\text{em}} = (3/2)^2 = 2.25$.

Canonical Prediction

$$\boxed{H_0^{\text{ORT}} = 67.4 \times 1.0859 = 73.19\,\mathrm{km\,s^{-1}\,Mpc^{-1}}}$$

SH0ES: 73.0 ± 1.0.
Deviation: $0.19\,\mathrm{km\,s^{-1}\,Mpc^{-1}} < 1\sigma$.
Zero free parameters.

16 Progressive Rendering Model

Hypothesis 16.1 (Redshift as Chain Growth)**.**

$$1 + z = \frac{|\mathcal{C}(t_{\text{now}})|}{|\mathcal{C}(t_{\text{em}})|}.$$

Conjecture 16.2 (Discrete Steps in $H(z)$)**.** $H(z)$ exhibits steps at:

$$z_1 \sim 0.35 \quad (\Delta H/H \sim 1\%), \tag{16.1}$$
$$z_2 \sim 1.1 \quad (\Delta H/H \sim 2\%), \tag{16.2}$$
$$z_3 \sim 2.8 \quad (\Delta H/H \sim 3\%). \tag{16.3}$$

For finite bins, observed positions shifted by $\Delta z \sim +0.3$.

17 Consensus Boundary Formalism

17.1 Definitions

Definition 17.1 (Consensus Boundary)**.** For observers $\mathcal{O}_1, \mathcal{O}_2$:

$$\partial\mathcal{C}_{12} := \{e \in \mathcal{U} \mid e \in \mathcal{C}_1 \triangle \mathcal{C}_2\}$$

(symmetric difference).

Definition 17.2 (Consensus Current)**.**

$$\boxed{J^\mu_{(C)} := \sum_i \rho_i(x) u^\mu_i}$$

where ρ_i is record density, u^μ_i is 4-velocity.

Definition 17.3 (Consensus Flux)**.**

$$\boxed{\Phi_C[\Sigma] := \int_\Sigma n_\mu J^\mu_{(C)} \sqrt{|g|}\, d^{d-1}x}$$

17.2 Conservation Law

Theorem 17.4 (Consensus Continuity)**.**

$$\boxed{\nabla_\mu J^\mu_{(C)} = S_C}$$

where S_C is source/sink of consensus.

17.3 Master Equation

Theorem 17.5 (Consensus Dynamics)**.**

$$\boxed{\partial_t \mathcal{R}(t) = \oint_{\partial\mathcal{R}(t)} J^\mu_{(C)}\, dS_\mu - \int_{\mathcal{R}(t)} \Gamma\, dV}$$

where $\Gamma = \Gamma_0 / N_O$ (FRQ decoherence rate).

Remark 17.6 (Physical Interpretation)*.* Growth of consensus reality = flux of new indexed events through boundary minus decay from decoherence.

17.4 Applications

Theorem 17.7 (Black Hole Horizon)**.** *For Schwarzschild horizon $r = r_S$:*

$$\Phi_C[r = r_S] = 4\pi r_S^2 \rho_0 u^T \neq 0$$

Information migrates to archival sector.

Theorem 17.8 (Bekenstein-Hawking Entropy)**.**

$$\boxed{S[\partial\mathcal{C}] = k_B \frac{A(\partial\mathcal{C})}{4\ell_P^2}}$$

Entropy of consensus boundary = Bekenstein-Hawking formula.

Theorem 17.9 (Cosmological Divergence)**.**

$$\boxed{D_C := \frac{|\Phi_C^{\text{late}} - \Phi_C^{\text{early}}|}{\Phi_C^{\text{early}}} = \exp(\alpha_0 \ln 2.25) - 1 = 8.59\%}$$

18 Numerical Verification: DESI Fire Test

18.1 Method

Computational suite (`scripts/desi_consensus_flow.py`) computes:

1. Consensus flux $\Phi_C(z)$ from DESI BAO.
2. Divergence $D_C = (\Phi_C^{\text{obs}} - \Phi_C^{\Lambda\text{CDM}})/\Phi_C^{\Lambda\text{CDM}}$.
3. χ^2 comparison.

18.2 DESI Year-1 Results

Table 4: DESI Year-1 consensus divergence

Quantity	Measured	ORT
$\langle D_C \rangle$	$8.2 \pm 1.5\%$	8.59%
χ^2_{red} (CDM)	2.16	—
χ^2_{red} (ORT)	1.44	—
$\Delta\chi^2$	3.6	—
Significance	1.9σ	—

Preliminary Confirmation

- ORT fits better than ΛCDM.
- Measured 8.2% vs predicted 8.59% (residual $< 0.5\%$).
- Year-3: expected 10–15σ per step.

18.3 Code Repository

`https://github.com/prtyboom/ontological-resolution-theory`
`scripts/desi_consensus_flow.py`

19 Big Bang as Anchor

Theorem 19.1 (Big Bang). *Big Bang is e_{BB} with $t(e_{\text{BB}}) = 0$, not moment $\mathcal{U}$ came into existence.*

Corollary 19.2 (Before Big Bang). *Question reformulates: which addresses in $\mathcal{U}$ lie outside $\mathcal{C}^{\text{cosmo}}_{\text{consensus}}$?*

20 Resolution of Paradoxes

Theorem 20.1 (Horizon Problem). *$\mathcal{U}$ static; all addresses connected. Inflation = rapid inclusion at early t.*

Theorem 20.2 (Flatness). *Curvature constrained by $\mathcal{P}(Z)$. Term $4\pi^2$ encodes 3D curvature.*

Theorem 20.3 (Singularity). *No ontological singularity in $\mathcal{U}$; only protocol boundary in $\mathcal{C}_{\text{consensus}}$.*

Theorem 20.4 (Information Paradox). *Information migrates across consensus boundaries. $\Phi_C[r_S] \neq 0$ implies no loss, only migration to archival sector.*

Part V

5D Archival Framework

Definition 20.5 (5D Metric).

$$ds^2 = e^{-2\Phi(z)/\varepsilon} g_{\mu\nu} dx^\mu dx^\nu + \varepsilon^2 dz^2$$

with $\varepsilon(N) = AN^{-\gamma}$.

Theorem 20.6 (Holographic Suppression).

$$\boxed{\varepsilon(N) = A\,N^{-\gamma}, \qquad A \approx 0.052, \quad \gamma \approx 0.57}$$

Theorem 20.7 (Effective Volume). *For $\Phi(z) = k|z|^p$:*

$$V_{\text{eff}}(\varepsilon) \propto \varepsilon^{1/p}$$

Theorem 20.8 (Dark Matter Ratio). *Matching $\Omega_{\text{DM}}/\Omega_b \approx 5.4$ yields:*

$$p = 1/\alpha = 0.59 \pm 0.02$$

Theorem 20.9 (Null Detection).

$$\sigma_{\text{arch}} \propto \varepsilon^{2p}$$

implying extreme suppression for $\varepsilon \sim 10^{-70}$.

Part VI

Finite-Rank Quantum

Axiom 20.1 (Finite Capacity). For every observer, $\exists N_O \in \mathbb{N}$: at most N_O distinguishable states accessible.

Corollary 20.10 (Decoherence Scaling).

$$\boxed{\Gamma = \Gamma_0/N_O}$$

Part VII

Verification and Falsification

21 Verification Suite

Test 1. $Z = 137.036046$

Test 2. $\alpha^{-1} = 137.035999$ (CODATA: 137.035999084)

Test 3. $m_p/m_e = 1836.15$ (exp: 1836.15267)

Test 4. $H_0 = 73.19\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (SH0ES: 73.0 ± 1.0)

Test 5. $D_C = 8.59\%$ (DESI: $8.2 \pm 1.5\%$)

Test 6. $(E_b/A)_{\max} = 8.821\,\mathrm{MeV}$ (^{62}Ni: 8.7945)

Test 7. $S_{\mathrm{BH}} = A/(4\ell_P^2)$ (derived)

Repository: https://github.com/prtyboom/ontological-resolution-theory

22 Falsification Criteria

ORT is falsified if:

(i) Vacancy-corrected α^{-1} excluded by improved measurements ($> 5\sigma$),

(ii) $H_0^{\mathrm{late}}/H_0^{\mathrm{early}} \neq \exp(\alpha_0 \ln 2.25)$ at high significance,

(iii) $H(z)$ steps absent in DESI Year-3 high-resolution bins ($> 5\sigma$),

(iv) FRQ scaling $\Gamma \propto N_O^{-1}$ experimentally excluded,

(v) Non-gravitational dark matter detection contradicts $\sigma \propto \varepsilon^{2p}$.

23 Open Problems

OP.1 Derive Z^2 exponent in top formula from protocol principles.

OP.2 Provide scheme-stable formulation of m_t (MC vs pole vs $\overline{\mathrm{MS}}$).

OP.3 Microscopic derivation of archival DOF scaling (A, γ, p).

OP.4 SNe Ia/BAO/CMB detailed fits for progressive rendering.

OP.5 Quantum gravity limit: derive Planck scale from $\mathcal{P}(Z)$.

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